Dev Minutes

Meeting: 22 September 2026

Time: 12:00 – 13:00

Attendance: All

AGENDA

1. **Introduction and process overview**
   - Explain how we plan to attack the project, agile/sprint approach, how decisions get recorded, what's expected of them between sessions
   - Tech Stack, react, javasscript … they are fine with ours as well.

2. **Concept walkthrough, section by section**
   - go through their note as written, confirming your understanding out loud before asking questions, so misreadings surface immediately.

3. **Clarifying questions**

Foundational / gap-filling

- Your note doesn't include a privacy/anonymity or synthetic-data section - what are your expectations there? Any specific thresholds in mind (e.g. minimum group size before a comparison is shown)?

- Students not less than 10 people, companies not less than 3
- If given result has one student, then add to another group
- Company grouping, size, venue capacity and operational scale, big medium large
- Anonymity – no student id, name or address, in data pipeline phase, also must randomise id
- restaurant and café must be differentiated (big, small, medium) venue capacity + operational scale (for grouping) more on size than the type

- Who's the actual end user at a business - owner, manager, staff

- 2 managers, 1 read only role extra e.g. staff member
- Admin roles to sign up for them (invite only through email)
- 3 roles (manager, read only for same vendor, admin role so that they can invite people to vendor role, and can change their password)
- To be clarified

- Are there different user roles? And if so - Do different users have different levels of access?)

- Yes

- Can one business have multiple users?
- Yes, have one account for multiple businesses (have drop down menu) don't create multiple accounts

- Can one user be associated with multiple businesses/locations?
- Split by pages

- Will BoschCard itself have access to the platform, or is it only for participating businesses? - new user role / premium user

- What is exactly meant by cafes / quick-service foods. Specifically cafes / coffee shops or just quick food in general
- café = quick food places – small meeting places for students

- Given six weeks, if only two of your four features can be a fully working MVP, which two matter most?
  - Non-negotiable = Rush hour radar + Demand Forecast

- Rush Hour Radar
  Hourly transaction heat map, eg 4pm periods, morning and lunch time rush hour periods.

- Privacy
  Suppress identification

- Demand Forecast
  Volume projects based on historical logs, forward estimates of customer demand, help with staff management

  Data Model
- Promotions for adding an event later with an extra column

Dashboard Expectations

- What should users see immediately after logging in?
- Main overview that contains rush hour radar, with tabs eg. Demand forecasting
- Analytics tab with depth
- Quick analytics
- Main overview – rush hour radar, tabs or buttons so you can switch to different screen like demand forecast, can filter more if you want later

- Want daily basis can go further ahead to see Friday (but want daily historical data) - decision support side of things, want to be able to discourage promotions if less rush hours), analytics tab that goes into the nitty gritty

- Which metrics are most important?
- What would make a business owner willing to pay for this service?
- What insights should stand out first?

Example: Should the dashboard show:

- Today's transactions
- Weekly trends
- Predicted demand
- Promotion performance
- Benchmark scores

Login/Registration

- Does the system need user registration?
- No
- Login?
- Yes
- Or should we assume the user already has access?
- By invite only
- Manager role, Read only role for vendor, One to invite
- Admin role should be able to send invite (email) to a role and then they can change their password themselves

Rush Hour Radar

- What granularity - hourly, half-hourly, day-parts like "lunch rush"?

- Days of the week and hourly
- Visual charts and heat maps

- How should "recurring" be defined - present in at least X of the last Y weeks?
- Should the promotional suggestion be a specific recommendation, or just a visual flag?

- What exactly should Rush Hour identify?

Example: Busy periods, Quiet periods, Both

- Both.
- If a threshold is met recommend a discount that would work, during quiet hours have promotions or discount to get people in and the opposite when it is busy, which is have no discount

- How should "busy" and "quiet" be determined? (Compared with the business's own historical activity, Based on a fixed threshold, Compared with similar businesses?
- Flag rush hour as quiet and busy

- How should busy periods be displayed? (Heatmap, Calendar, Line, heatmap, calendar, line graph, Hourly, hourly chart)

- How far back should historical analysis go (before pattern counts as reliable?) (Daily, Weekly, Monthly, Should the user be able to select a date range, Should the system consider day of week and time of day)

- What recommendations should be generated during quiet periods?
(Examples: Run promotion, Extend opening hours, Reduce staffing, Introduce specials)
- Discount amounts
- Discount amounts and promotions

- (Should these recommendations be automatically generated by the system, or should they be predefined rules based on certain conditions?) recurring quiet periods
- Automatically generated when it hits a threshold, is a recommendation (manager chooses whether to apply it or not)

Fair Comparison Score

- How should businesses be grouped for comparison by vendor type, size

- Vendor type + size

- Is the score a percentile, an index, an above/below-average label?

- Average, above below
- Identify general trend

-Heat map should also have vendor type

- Score percentile = specific screen for it, show how a venue compares to other venues of same size

- Compare cafes to the other cafes that are around you, aggregate every café and then can say whether they are above or below – have scale, use better language

- Minimum number of peer businesses before a comparison is shown?
- How should businesses be grouped?

For example: Coffee shops vs coffee shops / Fast-food vs fast-food or Businesses with similar transaction volume)

- Want small business to compare with each other and then within the café category, if everyone is having a bad week, we don't have to do anything because everyone else is also having a bad week

- What measurements determine a peer group?

Example: Revenue, Transaction count, Business category)

- Measurements to determine a peer group: can do a heatmap and say how many transactions they did, heat should be big to hide location, total + aggregate, value, heatmap should have a dropdown for vendor types

- How should comparison results be presented?

Example: Score out of 100, Percentile ranking, Better than average / worse than average

Honest Results Checker

- How should we handle exam periods, holidays and university events since these are not currently represented in the provided ERD?
- Can keep it out for now, but they will provide a calendar

- Where does exam/holiday/event data come from - do you have a calendar, or should we treat it as another synthetic feed? What counts as an "overlapping factor" worth flagging - do they want a fixed list (exams, holidays, events) or something more dynamic? Who maintains that calendar of events?

- Admin entering events, make predictions of that
- admin entering an event they know is happening (sat there is rugby so we know there will be more people, so they know from past)
- manager won't enter promotions
- contextual events - eng week, maties rugby, sa rugby, graduation, anything with a trend or event, add for context, pay day) dip in sales have to specify that there is holidays
- general student thing, not small clique of people

- How does a business indicate that a promotion occurred?

- Does the user manually enter promotions? (Even if they are automatically recommended?)
- No, they don't. They will just make a decision from the data

- What contextual events should be considered?

Examples: Exam periods, Recess periods, Maties events, Sporting events, Public holidays

- Sports, exams, etc
- Correlation is never causation
- Payday

- What should the warning message look like?

Example: "Sales increased by 18%, but exams and a university event occurred during the same period."

- Notification tab
- warning message: a notification bell to show to look at me - up to us how to say it. They said flags or pop ups would be annoying

Demand Forecast

- What horizon matters most - next day, week, month?

- Weekly

- Is a trend/seasonal-average model acceptable, or do you expect ML-level accuracy? (Worth calibrating now - it changes MVP scope significantly.)

- Leave out ML for now

- How far into the future should forecasts predict? And what should it be predicting?

Examples: Next day, Next week, Next month

- What output should be shown?

Examples: Expected transactions, Expected revenue, Expected customer count

- 1 graph with 3 different lines

- How much forecasting accuracy is expected?

Since you're using synthetic data, this helps define scope.

- Deferred

4. **Final Product Expectations**

- What does the client consider a successful product?
- Reliability, private, match requirements, represented in a clear manner, understandable,
- Reliable, private, database constraints to match requirement given, presented in clear manner, visual appealing, easy to read, vendors can make better informed decisions comparing to what other people are doing

- Should the system be a web application?
- Yes

- Should users be able to export/download reports?
- Typst, yes
- Make something where you can make/export weekly reports like typst where you can download weekly information

- Should the system provide recommendations automatically?

- Should there be notifications/alerts?

- Should users be able to interact with charts and filters?

- Yes, details through hovering (built in apparently)

5. **Wrap-up**
   - confirm deliverables for them before Session 2 (e.g. "bring examples of a report you'd want to see")
   - BRD, UML, Use Cases, User stories
   - 3 high-fidelity Wireframes, Dashboard with analytics, Color palette choice
   - Login , Dashboard, Analytics

Tech stack direction

No stack is mandated - but it must be justified in writing, and needs to run against the given ERD via a real database (not a spreadsheet). Given the ERD (Student, Vendor, Vendor_Type, Transaction - a simple relational structure) and that this needs to be buildable and explainable by a small group in ~ 6 weeks:

- **Backend + DB:** PostgreSQL fits the ERD cleanly (proper foreign keys, easy joins for the aggregation logic Fair Comparison Score and Rush Hour Radar need). Pair it with a beginner-friendly backend framework - Python + FastAPI is a common, well-documented choice with a gentle learning curve if your group has mixed experience.
- **Frontend:** keep it simple a basic dashboard (even a lightweight React or plain HTML/JS front end) is enough; the marks are on working functionality and justified choices, not visual polish.
- **Justify by:** team's existing skills, PostgreSQL's fit for the relational ERD, FastAPI/Flask's low setup overhead, and the project's fixed timeframe.

Data for better decisions not intuition.

7. Open Questions for Session 2

- Definition of "recurring" (X of last Y weeks)
- The three series on the Demand Forecast graph
- Whether BoschCard itself gets a platform role
- Do you still want the admin to log contextual events
- What are the actual differences between a manager and a staff member in roles = what exactly is being edited by the manager cause the whole system is already supposed to be read only
- Honest Results checker how would you want it to be represented? Notification bar / bell or hover feature on the demand forecast?
- Ask about the calendar. Cannot add any new/created data that does not fit the model
- For Honest Results Checker: Is it only going to be a notification bell and then you click on it and it shows you the message OR incorporate the honest results checker into every principal feature and when you hover over a graph it will show the message. Example: There was a promotion. There was a Maties Rugby event on the same day. Sales increased by 15% on this day.
- What type of heatmap is wanted? An individualized heatmap for specific vendor based on the user/heatmap with map of Stellenbosch general heat for all Venue types/ heat map filtered by venue type
- NB! Ask them to send the synthetic data to us

Byte & Bite - Session 2 Prep

Date: 29 September | Time: 10:30 | Format: In-person | Attendance: All

AGENDA

1. Walk through the BRD section by section, confirm/decide as we go
2. Work through the open questions below
3. Get the synthetic dataset from them priority #1, this blocks development
4. ROLES & PERMISSIONS

Foundational question first: There are three roles (Admin, Manager, Staff) confirm exactly what distinguishes them. Specifically: what can a Manager actually edit that a Staff member can't, given the system is otherwise read-only for everyone? If nothing is genuinely editable by Manager beyond viewing, "Manager vs Staff" may really just be a visibility distinction (which screens/data each sees), not an edit-permission one worth naming that explicitly rather than assuming an edit capability exists.

Already decided (our call, informing them):

- Staff will not have access to report exports (Manager/Admin only)
- Multi-location access will be a dropdown switcher, not a combined view

Rows still needing confirmation:

Admin role as byte and byte will see all the vendors that subscribe to our system
invite sent to users when they subscribe to our services + load their email
Try and attempt to invite other managers + from their specific vendor

Vendor side : Manager

1. **Fair Comparison Score - Staff**

Question: Should Staff have no access, or a simplified version without exact figures?

Build: Backend returns 403 for Staff role; frontend hides the tab entirely.

Want to have it hidden

2. **Demand Forecast - revenue line for Staff**

Question: Full access, or drop the revenue line and keep only transactions/customer count?

We decided to drop the revenue line and keep transactions and customer count

Admin = full access of everything

Build: Backend strips the expected_revenue field from the response for Staff before sending enforced server-side, not just hidden in the UI.

3. **Honest Results Checker — Staff scope**

Question: Contextual notes only on traffic-related data Staff already sees, or also on revenue-tied notes?

Staff should not see revenue, staff should only see contextual notes

Build: Falls out automatically from #2 - a note explaining hidden revenue data is never sent to Staff either.

4. **Vendor profile settings — Manager**

Question: Can Managers edit (name, address, category), or is it view-only with Admin as the only editor?

Admin is the only one that can edit the details. Manager will send their info to admin if they want name changed.

Build: If yes - edit form rendered for Admin and Manager, backend checks role on save. If no - same screen, fields disabled for Manager.

Security note to state plainly to the client: role checks happen on the backend, not just by hiding buttons a Staff member can't retrieve restricted data even by inspecting network requests directly, since the server itself withholds it.

2. **DEMAND FORECAST**

Question: The notes say "1 graph, 3 lines" - confirm the exact three metrics. Current assumption: Expected Transactions, Expected Revenue, Expected Customer Count. Should any of these be swapped or supplemented (e.g. a confidence interval band, or last week's baseline for comparison)?

Also confirm: since ML is out of scope, are they expecting a specific heuristic approach (e.g. trend/seasonal-average), or is that our call to propose?

Question: Can we add another line to the demand forecast Average order value vs Average spend per student
AOV = Total revenue / Total transaction count Or ASPS = total rev / anonymised student count (fewer than 10 students = mask data)

Average spend per student ? =
want to keep the 3 lines still === undecided bout it

Will come back to us.

3. **RUSH HOUR RADAR**

Question - heatmap scope: which of these three did they actually mean?

- An individualized heatmap for the logged-in vendor only
- A general heatmap across all of Stellenbosch, all venue types combined
- A heatmap filterable by venue type

Want a geographical heatmap filterable by venue type
Heat map for different vendor sizes filter
Want to use the maps as priority - general heatmap across all of Stellenbosch

For example, will have a zone, example 5 km and then all of the vendors in that zone will be either show busy or not (so zones will be busy or quiet include all of the vendors in that zone)

operational size filter by transaction / customers (will then have a group either small, medium or large based off of their transactions)

Use filters

Colour gradient

Question - "recurring" definition: what's the actual rule? E.g. a quiet period must occur in at least 3 of the last 4 same-weekday occurrences before it's flagged.

Quiet period = 30% below weekly baseline + 20% above the weekly baseline

More categories: quiet, moderate, busy and very busy = 25, 50, 75

4 weeks baseline on the same day/hour

We can decide the percentiles (25, 50, 75 – give the labels don't show the percentages)

Question - recommendation logic: when a recurring quiet period is found, should the suggested discount be a fixed tier (e.g. "10-15% between 15:00-16:30"), or dynamically based on how past promotions performed?

Dynamic based on past discount performances

4. **HONEST RESULTS CHECKER**

Question: Is this a notification bell the user clicks to see a message list, or should context be embedded directly into every feature - e.g. hovering over a graph point shows "Sales increased 15% - a promotion was active and there was a Maties Rugby event the same day"?

Both mechanisms together

Notification bell:

When notification clicked = show notification list + when clicked = have a detailed breakdown NOO CAUSATION + only flag that the 2 events coincided

Notification bell - automated alert system - when system has a notable shift in sales or etc, then you can click on it and then you will see unusual transaction volumes which overlaps with rugby event

Hover:

Hovers include = contextual detail on graph + on data point on graph = see some data + see detailed data on workflows

5. **CALENDAR & CONTEXTUAL EVENTS**

Question - data source: confirm whether we preload a static academic/sports calendar as seed data, while leaving the Admin UI open for ad-hoc additions (payday, test weeks, campus events) - or whether they're providing the full calendar themselves.

Question - category structure: does Admin pick from a fixed list (exams, holidays, sport, payday), or is it free-text entry? Determines whether we build a dropdown or an open field.

Admin should be able to create, update, delete - they were not sure here actually

Constraint to state clearly: we cannot add new fields/entities that don't fit the provided ERD - any calendar data will need to live in a small supporting table separate from the four required entities, not bolted onto them.

6. **OTHER OPEN ITEMS**

- BoschCard institutional role: does BoschCard itself get a separate super-user tier for macro-level campus insight, or do they just use a standard Admin account?
- Weekly report (Typst) contents: what sections must be included - e.g. weekly peak/lull summary, forecast accuracy variance, Fair Comparison standing?
- Vendor "size" / operational scale: not in the ERD - confirm whether it's computed dynamically from transaction volume, or manually set by Admin per vendor.
  Are okay with using transaction volume
- Location precision: exact GPS/address stored, or zone-level from the start (ties into their own stated privacy requirement)?
  Remove some of the coordinates

BEFORE WE LEAVE TODAY

Get the synthetic dataset from them - or a clear timeline for when we'll receive it. Nothing else on this list can be built against real structure without it

Byte & Bite - Session 3 Prep

Date: 6th October | Time: 10:00 | Format: In-person | Attendance: All

Feedback for mvp version 1:

Map

- Try get scrolling, two finger gestures for moving
- Dots too small, make the area / dots bigger
- Increase the font on the demand forecast one
- Lighter logo or better contrast, side tab and logo
- Top right logs out
- Land on the dashboard, should be a quick glance
- Dashboard?
- Trigger occurs when a threshold appears, 30% increase/decrease for notification
- Notification pops up if it's a regular occurence it can be on a different page
- Pin notification and a clear option for notification
- Filter by day for fair comparison
- Drop threshold to 5 students

Rush Hour Radar Options :

The issue: Our privacy rule says we need at least 10 different students in a time slot before we'll show any data for it - anything below that stays hidden, to protect individual students from being identifiable. When we tested this against real data, most 3-hour meal-time slots only have 3-9 students, so almost everything was staying hidden. Here are three ways to handle that:

All based on 3-hour time slots

Option 1 - Lower the threshold to 5

- **What it means:** We show data once 5+ different students are in a time slot, instead of waiting for 10.
- **Advantage:** Most vendors show real, useful data most of the time.
- **Tradeoff:** Slightly less privacy protection a group of 5 people is a bit easier to narrow down to individuals than a group of 10.
- Drop it to 5 students

Option 2 - Lower the threshold to 3

- **What it means:** We show data once just 3 different students are in a time slot.
- **Advantage:** Almost every vendor shows data, almost all the time.
- **Tradeoff:** This is the weakest privacy protection of the three - a group of 3 is genuinely small enough that it could risk identifying who was there.

Option 3 - Keep the threshold at 10, but look at more weeks of history

- **What it means:** Instead of asking "were there 10 students in this exact 3-hour slot today," we ask "were there 10 students in this slot across the last 10 weeks combined."
- **Advantage:** Keeps your original 10-student privacy rule completely intact nothing is weakened.
- **Tradeoff:** The data shown reflects a longer-term pattern rather than just this one day, so it's slightly less "in the moment" - though realistically, a longer trend is often more reliable anyway.

One line to close with, if helpful: "Our recommendation is Option 3, since it doesn't touch the privacy number you gave us - but we wanted you to see all three and make the call yourselves."

- The main Rush Hour Radar is a geographical map of Stellenbosch divided into zones, where each zone is classified as quiet, moderate, busy or very busy based on the vendors within that zone. Users can filter by vendor type and operational size. Is this correct?

Log In:

- Forgot password: What should happen when a user clicks "Forgot password"? We have 2 different options: Should they receive a password-reset email, or should the Admin reset their password?
- Take you to a different page, then ask you to log in. We can say we sent an e-mail and took them to this page
- Initial page: After logging in, should the user land on the Dashboard/Rush Hour Radar, or another page?
- Invite: When an Admin invites a user, should the invitation contain a link where the user creates their own password?
- Admin invites user, user taken to link this is your email and set password, given 24 hours
- Use gmails , and find out about email services
- Failed login: What should happen after an incorrect email/password? Just a simple error message, or should there be an account lockout after multiple attempts?
- Pops up message to admin, either revoke access or admin sends a reset password, incorrect e-mail or password
    - Forgot password? → email reset link or Forgot password? → Admin resets it.

Demand Forecast: Show them how it looks – then show wireframes to be better

- Forecast period: Should the forecast always show the next 7 days from the latest available transaction data, or should it always show the next 7 calendar days from the actual current date?
- Three lines: Can we confirm that the three lines remain:

1. Expected Transactions
2. Expected Customers
3. Expected Revenue
4. **Average spend -** Do you still want Average Spend per Student/Customer included, or should we keep the graph to exactly three lines?

- Hover: When hovering over a point, what information should be shown? Just the three values, or contextual information as well? (calendar events)

More questions (demand forecast)

- Hover function – what exactly should pop up when a point is hovered over

- Currently we have a drop down for the daily figures?.....(Let them look at it) Would you prefer this to be a wireframe or is this look we have

Fair Comparison Score

- When a manager opens Fair Comparison, what do you want them to see first?

For example: **Your business** - Transactions: 1250, Peer average: 1050, Result: Above average

And Then: **Peer group** - Small cafés, 8 participating businesses, Average transaction volume

- Should the comparison show the actual numerical peer average, or only say Above Average / Below Average?" - recommended = show **aggregated statistics**, not individual business values.

Honest Results Checker:

- What should trigger an Honest Results notification? Should we define a threshold ourselves, for example a significant increase/decrease compared with the normal baseline?
- Should notifications remain in the notification list after they've been viewed, or disappear once read?

Contextual Events / Calendar:

- Should Admin choose an event type from a predefined list, such as Exam, Holiday, Sport, Payday, Graduation, etc., or should Admin be able to type their own event?
    - recommend: **Event Type dropdown + Event Name + Date**

Reporting:

- What information should the weekly report contain?
- Should there be an export button on each of the 3 features pages and can only export info from those specific pages or across multiple features?
- Given discretion

Dashboard:

- Do you want a dashboard or once you log in the first thing you see is the rush hour-radar?
- If there is a dashboard, what do you want on it? - be specific eg peak days /
- Given discretion

Callender clarification

What we're actually building

A list of important dates - exam periods, holidays, graduation, recess, rugby matches - that the system can check against when something unusual happens in the sales data.

Why we need it

If sales suddenly jump or drop, the system shouldn't just say "a promotion worked" or "something's wrong" - it should check: *was anything else going on that day?* If yes, it says so, instead of letting the business owner draw the wrong conclusion.

How it works, step by step

1. **We have a real list of 191 Stellenbosch dates** - already verified, not made up - covering things like exam weeks, recess, public holidays, graduation, orientation week, and major events.
2. **We keep this list in a simple file, not a database table.** This matters because our lecturer controls the exact data structure (the ERD) our database is allowed to have and it's only meant to hold student, vendor, and transaction records. The calendar doesn't belong in there, so we keep it separate, as its own file the app reads.
3. **The system checks for overlaps.** When a manager looks at a sales change, the system quietly checks: *does this date range match anything on our list?* If yes, it shows a note something like "Sales rose 15% this week note: this overlapped with Graduation." It never claims the event *caused* the change, only that they happened at the same time.
4. **Admin can update the list.** Instead of adding events one at a time through a form, the Admin uploads a file (like a spreadsheet) with new dates, and the system swaps in the

updated list. Less flashy than a typical "add event" button, but fully functional and it avoids touching our required database structure.

The one-sentence version for your clients

"We're keeping your calendar of important dates as a simple, uploadable list rather than part of the core database so your Admin can update it whenever needed, without us having to change the underlying system."

- Create database tables , it doesn't necessarily have to be boschcard data

- Less technical questions
- What is the shape of the data...

- Rush radar - Extension of the radar = giving actual steps / methods to get to a highpoint

Manger should be able to see how they did last week where we did well = code something into factors that we can implement this week to get good results

Agenda 1: Team prep meeting (before the client session)

Byte & Bite: Team Prep. Tuesday 13 October | Time: 11:00 | Attendance: all

1. Opening (5 min)

- Goal: be demo-ready and agree who says what at 12:00.
- Where we are against the timeline: second MVP stage, privacy and Fair Comparison.

2. Individual updates (about 5 min each)

Member	Worked on
David	Fair Comparison backend, report feature
Nicole	Fair Comparison front end, consistent tabs
Victoria	Calendar, front-end tweaks, dashboard, merging into main
Matthew	User access (login, invites)

Questions for individual items

David

Nicole

Victoria

Matthew

3. Demo readiness (10 min)

- Does the demo run end to end on the machine we'll present from?
- Which pages are we showing: Rush Hour Radar, Demand Forecast, Fair Comparison, dashboard, calendar?
- What is not finished, and how do we explain it simply?

4. Decisions to settle internally (10 min)

- Who owns Honest Results Checker, or do we show it as "coming next"?
- Which email service for invites and resets? The client said Gmail, so do we just confirm that?

5. Roles in the client session (5 min)

- Chair, scribe, who presents each feature, and who handles questions. - Matthew
- Rule: no technical words with the client. Describe what they will see and do.

7. Other

Agenda 2: Client session

Byte & Bite: Session 4 with Group 4. Tuesday 13 October, 12:00 | In person

1. Welcome and recap (5 min)

- What we heard last time and what we changed: bigger map dots, better contrast, a quick-glance dashboard, pin and clear for notifications, the day filter for Fair Comparison, and the lower student threshold of 5.
- Please think of other things to add to recap

What changes with the data
- double data, double students, double the niche companies, closer to the erd

2. Demo (about 20 min)

- General Layout
- Tabs should stay in the same spot when you scroll on the page
- Make all the filters be in the same position/spot for each page
- New data has more data for restaurants and cafes for our niche and the layout is the same

- Rush Hour Radar - Scrolling of the map (don't show)
- Change to a more readable table format
- Arrange vendors, sort by ...
- Make sidebar fixed
- Add pagination for 15/20 etc
- Have colour status go to the bottom part
- Make it know that you can't click on the button (for holding price and discount)
- Have filter for the holding price and discount so that you can see who needs it the most (put discount above holding price)
- For the holding prices and discount part have number of pages (1 to 15) rather than a continuous scroll and then more pages with a next button

- Login and access - how to invite user as well
- Login as staff member, demand forecast (2 lines) and rush-hour radar

- Login as an admin - show features
- When admin adds events , eg fair comparison score they should see an event happened

- Demand Forecast (Export feature for reports)
- Make headings bigger on the cards

- Want a search bar for the dropdown so that they can type
- Maybe add a heading for vendor
- Add dbounce to search bars
- Add faint grid pattern for the export function
- Also, faint grid on demand forecast and have it also be on the pdf export
- Don't need dropdown for the weekly demand at the bottom of the page
- Want title for vendor for search bar to say vendor
- Extra feature falls into honest results checker
- Honest results checker = context
- Extra feature = add a recommendation, so suggest discount or event to get more people in
- For single date want something that adds more value rather than just filling space for line graph

- Reports
- Staff members should be able to export reports, but should not be able to see revenue
- Want to be able to access the past downloaded report
- Manager should be able to see who has exported what and when
- Should be able to export reports for each page

- Fair Comparison Score (David and Nicole)
- One day, bar graph
- If its one day , add more categories to make data more manageable
- For specific day make it a bar graph
- When admin adds event

Not sure what page specifically but:

- On one of the pages, user should be able to look at past performance and look at a specific date be able to see what event was happening on that day and can see that is what happened on that day

- Calendar of events (Victoria)
- Add an actual calendar
- Add a clickable feature that you can view it as a calendar, and then also view it how it is now

- Dashboard (for next meeting)
- The first one, change font, make it easier to use - can use font from second picture on first layout

- Demand forecast, add day of the week eg Monday 29 December. Top blocks you don't a little icon, strip with different colours emphasize numbers
- Hover on widgets: message “click to go to feature page “
- Heatmap needs more contrast
- Simplify filters, remove one date, and multiple vendors
- Layout, horizontal and vertical
- Fix cards to be more uniform
- Filters can just float no background needed
- Make the cards on the top have more emphasis
- Make the links for each feature (widget) make it show that it can go to another page that it is for (make user know that it is clickable and where you will go)
- Make the horizontal spaces and the vertical spaces between cards and line graphs the same

Try maintaining layout on each page, especially for filters as well.

Staff member

Demand forecast

- Vendor name aligned

3. Fair Comparison Score: questions for the client (10 min)

1. When a manager opens this page, is the first thing they see what you expected?
2. Do “Above average” and the colours make sense at a glance, or would you want different wording?
3. Is “similar cafés of the same size” the right way to describe the group they are compared with?
4. Should the page show the actual average number, or only whether they’re above or below?
5. Is the day filter (for example, Mondays only) easy to find? Would you want other filters?
6. Is the trend line useful, or does it make the page feel busy?
7. If there aren’t enough similar businesses to compare, is a short “not enough data yet” message acceptable?

Report Feature

Questions for the client about reports

1. **Contents:** when a manager exports a report, what should be in it? Just the page they're on (for example the forecast), or one combined weekly summary?
2. **Who gets it:** is the report mainly for the manager to keep, or to share with someone else, like the owner or a team?
3. **Format:** is a PDF what you'd expect, or would you also want something like Excel?
4. **Look:** here's an example of an exported report. Does the layout work for you? Would you add your logo, your business name or a date range?
5. **History:** should the Reports page show a list of everything that's been exported before? Would you want to re-download an old report, or is a list of what was exported enough?
6. **How often:** do managers need reports weekly, daily, or only when they ask?

4. Other question features from everyone

5. Other questions for the client (10 min)

- **Dashboard:** what should a manager see in the first 10 seconds after logging in?
  Decided by Victoria, show screen and ask if they want anything different
- **Weekly report:** what should it contain, and is one report per page right, or one combined report?
- **Calendar events:** is a dropdown of event types (exam, holiday, sport, payday, graduation) plus a name and date enough?
  Calendar stuff still to be decided
- **Notifications:** how long should they stay in the list after being read?
- **Forgot password:** is an email link fine, or should the Admin reset it?
  Login related

6. Next steps and feedback (5 min)

- Recap what we agreed.
- Confirm what we will show at the next session on 20 October.

Agenda

Byte & Bite: Session 5 with Group 4. Tuesday 13 October, 12:00 | In person

In Session 4 we reviewed the pages built so far and agreed on a round of improvements. The Dashboard was to become a clearer overview, with quick stats and widgets that link to the other pages, and a more consistent look: uniform cards, equal spacing, and filters and tab positions that stay the same across pages. For Demand Forecast we wanted bigger card headings, a vendor search, and a weekday name on its dashboard widget. For Fair Comparison we asked for a single-day bar graph. For Rush Hour Radar we asked for a sortable, filterable table with pagination.

We will go through each feature and tab and get your feedback.

Side bar collapsable, keep icons and icons only, you can hover on the icon

Need to e-mail Josh about what we want the data to look like

Login

Invite user

- **Side tab**
    - **Forgotten password – auto make it so user can reset their own password, send email get link change password, admin can still reset password and there is an audit log of who changed password**

Create User

- **Do you want it as a tab only in admin profile or its okay at the top**
- **To reset password, you would go back to create password**

Steps for creating user = Take time on each one of the screens
1. Login as Admin user
2. Go to top right click on admin role
3. Click on users
- Do they like the placement of the users tab + how it appears on the tab after it is clicked for navigation ?
4. fill out the user details and invite a user to the system
    - ask clients how they feel about the look and feel how the screen is being displayed? Do they have any submissions / Any changes ?
5. Log out as an admin
6. Click on create user with token
7. Copy and paste token from terminal where you are running your backend

8. Go back to screen and set password
9. Login with new user.

Forgot password:

1. Manager wants to change password
2. Click forgotten password
3. Sends notification to admin
4. Admin clicks rest password on their page
5. That should send an email link to the manager
6. They click on email and then they click copy token from email
7. Create account / password then input the token and set new password

Forgot password

They don't want it to go to the admin

Dashboard

- Bar at the top with start dates need to go
- Change dashes by the time area
- Still remove background at the top
- Demand trend should have the day of the week

Rush Hour Radar

- We can remove the dismiss option on the auto-generated

Rush-hour radar Recommendation

- When you click autogenerated recommendation, when you click a specific vendor will you take you to the dashboard, admin or manager.
- Go to vendor
- TBD who can view the names of company
- Admin allowed to see the company information
- Manager and staff will only see their own information and but the company colors
- Recommendation in a table format with different columns and searchable
- Be able to go to demand forecast
- Last updated recommended – time it was generated
- Filter for status and search box on the rush hour radar side bar with companies, you can also sort them

Demand Forecast

- Extra feature is this what you wanted, demand recommendations

- Exporting report, fix it
- Expected revenue e.g. make two decimals, make the comma a.
- Recommendations change the way it looks, sort it or filter (make it a table)
- Can they check the recommendation and verify by going back to the previous day,
- Donut graph for the 30 %, different colours for the different recommendations, half the card on the demand recommendations
- Change colours for the high demand, above average
- Demand recommendation, calculate what percentage or discount to try match performance

Notifications

- If there is more than one notification, show small preview stacked with drop down option

Demand Forecast and Honest Results Checker

- Centre notch in the middle of the page change colors to make them more noticable, thinner bar rounded edges

Honest Results Checker

- Find a better way to display information
- Remove curly brackets

Fair Comparison

- Added more context
- Specific periods added

Calendar/Context Events

- **First card fix display eg. Start date etc, layout**

Reports

Settings tab

- **What would you want in your settings**
- **Scrap**

1. Quick status round (about 10 min)
One line each on done, in progress and blocked, in this order: David, Matthew, Victoria, Nicole.

2. David: Dashboard and Reports

    - Report export on all tabs: Dashboard, Rush Hour Radar, Demand Forecast, Fair Comparison and Calendar.
    - A Reports tab that logs who exported what and when.
    - A Dashboard redesign: coloured strips on the KPI cards, a layout aligned with the charts, a faint grid on the trend chart, and widgets that link to their pages with a hover hint.
- **To demo:** export a Dashboard report, then show it in the Reports tab.
- **Still open from Session 4:**
    - Heatmap contrast.
    - Simplified filters, floating without a background.
    - Weekday name on the Demand Forecast widget.
    - The font change (I still need to know which font was meant).
- **Decisions needed:**
    - Add a preview column to the report exports table to store a snapshot of each report. This is a database change, so it needs Victoria's agreement.
    - Preview logging users out: is the cause a new tab or an expired token?

3. Matthew: Login and Access

- Status of login, roles and user creation.
- Staff and Manager test accounts with linked businesses, which I need to test the role rules.
- Later: the synchronised page layout per role.

4. Victoria: Calendar and Honest Results Checker

- Calendar view toggle, and whether the print layout is fine.
- Honest Results Checker: the recommendations item from Session 4.
- Merging branches into main.

5. Nicole: Rush Hour Radar, Demand Forecast, Fair Comparison

- **Rush Hour Radar:** table, pagination, sort and filters.
- **Demand Forecast:** bigger card headings, vendor search with debounce, "Vendor" label, faint grid on the chart and PDF, remove the weekly dropdown, and a readable grid.
- **Fair Comparison:** a bar graph for a single day, and y-axis numbers on the trend chart.

Minutes

Demand forcast

- **Space out amounts - in the thousands**
- **Make all date formats the same**
- **Make day and date column separate**
- **Match change and activity category**
  Historical Activity Change
  - **Filter that decides between transaction count and revenue count**
  - **Take away the hover option**

Rush-hour radar

- Auto generated recommendation change view so that its more table format like the demand forecast

All figures - must have two decimals

Look into loading time

- Put skeletons - form the initial cards
- Cache calculations
- Device detected preloading
- What do we need to fetch for each feature

Changes to login

Dashboard

- If busiest period doesn't show say why

Vendor types

1 Restaurants

2 Cafe'

3 Bakery

Colour scheme

- logo make it a cookie colour

Mailpit for emails

Read through concept brief

Final ENV File

```
DATABASE_URL=postgresql://neondb_owner:npg_hf9cAO6Lgikn@ep-dry-boat-b52jw2ps-pooler.c-7.us-east-2.aws.neon.tech/neondb?sslmode=require&channel_binding=require
JWT_SECRET=OBpbrJ7fEUriRLwIBGHtJC78eg03UEe33eQz9pTC_CBhGHEhrU9uHTajMBMuseZ_
FRONTEND_URL=http://localhost:5173/
GMAIL_ADDRESS=cokergk199@gmail.com
GMAIL_APP_PASSWORD=lixggtnuxwlssajk
```

To Do List for Friday

NB!

- Fix up and clean each file especially the app.css, add/change comments and check AI use

Data - Victoria

- Filter and clean data and import it,
- Make sure that everything works (if things don't work then we will all help)

Other Documents

- SCARF (done individually)

Layout - Nicole

- Make filters all the same layout
- Make Export button the same size as one another

How it should look:

- (maybe make the dropdown look like fair comparison, but make fair comparison searchable and look like demand forecast)
- Rush hour radar search bar dropdown layout should look like the fair comparison drop down layout
- Size and layout of the filters on the rush hour radar should look like demand forecast
- Make demand forecast and rush hour radar cards look the same
- Remove refresh button on dashboard page and reports page

General - David

- Do loading speed for each page and database

Rush Hour Radar - Matthew

- Should the zones be made bigger (check after data has been added if they need to be made bigger for privacy reasons)
- Make managers/staff should not be able to click on zones in heatmap and see other businesses information
- Managers/Staff should only see their business in the vendor pulse (on right hand side of heatmap) and only their auto-generated recommendations

Demand Forecast - Matthew

- Should open to the latest date so that it can forecast the next week (can only check this once data has been added)
- For the export button in the pdf the activity category gets cut off need to fix that

Reports - Nicole

- Fix the css on this

Notifications - David

- Where should each notification bell on each page take you – answer: (have it so that when you click notifications on the other tabs it shows what it would show on the demand forecast)
- Check why there are duplicates on view all notifications